

NUMERICAL METHODS

Unit-4: Numerical Differentiation and Integration

Comprehensive Study Notes



4.1 Introduction to Numerical Differentiation

Numerical Differentiation is the process of finding the derivative of a function using numerical techniques when the function is given in tabular form or is difficult to differentiate analytically.

Why Numerical Differentiation?

- Function is given ONLY as a table of values
- Function is too complex to differentiate analytically
- Need quick approximation of derivative
- Analytical derivative may not exist

4.2 Differentiating a Graphical Function

When a function is given as a graph, find derivative by drawing a **tangent** at the desired point.

Steps for Graphical Differentiation

1. Draw the graph of $f(x)$ accurately
2. Identify the point x_0
3. Draw tangent at $(x_0, f(x_0))$
4. Extend tangent to form a right triangle
5. Measure Δy (vertical change) and Δx (horizontal change)
6. $f'(x_0) = \Delta y / \Delta x$
7. Record the approximate derivative

Advantages and Disadvantages

Advantages	Disadvantages
Visual understanding of derivative	Less accurate than numerical methods
Quick estimation	Subjective - depends on person drawing
No complex calculations needed	Depends on graph scale
Good for preliminary analysis	Only rough approximation

4.3 Differentiating a Tabulated Function

When a function is given in **tabular form**, use finite difference formulas.

4.3.1 Equal Interval Formulas

For equally spaced data with step size $h = x_{i+1} - x_i$:

Forward Difference Formula (At Beginning of Table)

$$f'(x_0) = (1/h) [\Delta y_0 - (1/2)\Delta^2 y_0 + (1/3)\Delta^3 y_0 - \dots]$$

Backward Difference Formula (At End of Table)

$$f'(x_n) = (1/h) [\nabla y_n + (1/2)\nabla^2 y_n + (1/3)\nabla^3 y_n + \dots]$$

Central Difference Formula (At Interior Point)

$$f'(x_i) = (y_{i+1} - y_{i-1}) / (2h)$$

Second Derivative Formula

$$f''(x_i) = (y_{i-1} - 2y_i + y_{i+1}) / h^2$$

Example: Find $f'(1)$ from given data

x	0	1	2	3	4
y	0	1	4	9	16

Step 1: Construct Difference Table

x | y | Δy | $\Delta^2 y$ | $\Delta^3 y$

0	0			
		1		
1	1		2	
		3		0
2	4		2	
		5		
3	9		2	
		7		
4	16			

Step 2: Apply Forward Formula at $x_0 = 1$

$h = 1$

$\Delta y_0 = 3, \Delta^2 y_0 = 2, \Delta^3 y_0 = 0$

$$\begin{aligned}
 f'(1) &= (1/1)[\Delta y_0 - (1/2)\Delta^2 y_0 + (1/3)\Delta^3 y_0] \\
 &= 3 - (1/2)(2) + 0 \\
 &= 3 - 1 = 2
 \end{aligned}$$

Actual: $f(x) = x^2, f'(x) = 2x, f'(1) = 2$

Example: Second Derivative

Find $f''(2)$ from the same data:

$$\begin{aligned}
 f''(2) &= (y_1 - 2y_2 + y_3) / h^2 \\
 &= (1 - 2(4) + 9) / 1 \\
 &= (1 - 8 + 9) \\
 &= 2
 \end{aligned}$$

Actual: $f''(x) = 2, f''(2) = 2$

4.4 Introduction to Numerical Integration

Numerical Integration is the process of finding the definite integral using numerical techniques.

$$\int_a^b f(x) dx = \text{Area under the curve from } a \text{ to } b$$

Newton-Cotes Formulas

Divide interval $[a, b]$ into n equal parts with step size $h = (b - a) / n$:

n	Method
n = 1	Trapezoidal Rule
n = 2	Simpson's 1/3 Rule
n = 3	Simpson's 3/8 Rule
n = 6	Weddle's Rule

4.5 Trapezoidal Rule

Approximates the area under a curve by dividing it into **trapezoids**.

Formula

$$\int_a^b f(x) dx = (h/2) [(y_0 + y_n) + 2(y_1 + y_2 + \dots + y_{n-1})]$$

where $h = (b - a) / n$ and n = number of subintervals.

Example: Find $\int_0^4 x^2 dx$

x	0	1	2	3	4
$y = x^2$	0	1	4	9	16

$$a = 0, b = 4, n = 4$$

$$h = (4 - 0) / 4 = 1$$

$$y_0 = 0, y_1 = 1, y_2 = 4, y_3 = 9, y_4 = 16$$

$$\begin{aligned} \int_0^4 x^2 dx &= (1/2) [(0 + 16) + 2(1 + 4 + 9)] \\ &= 0.5 [16 + 2(14)] \\ &= 0.5 [16 + 28] \\ &= 0.5 \times 44 \\ &= 22 \end{aligned}$$

$$\text{Actual: } \int_0^4 x^2 dx = [x^3/3]_0^4 = 64/3 = 21.333$$

$$\text{Error} = 22 - 21.333 = 0.667$$

4.6 Simpson's 1/3 Rule

Uses **parabolic approximation** for better accuracy. **Condition: n must be EVEN.**

Formula

$$\int_a^b f(x) dx = (h/3) [(y_0 + y_n) + 4(y_1 + y_3 + \dots + y_{n-1}) + 2(y_2 + y_4 + \dots + y_{n-2})]$$

Example: Find $\int_0^4 x^2 dx$

$$a = 0, b = 4, n = 4 \text{ (even)}$$

$$h = (4 - 0) / 4 = 1$$

$$y_0 = 0, y_1 = 1, y_2 = 4, y_3 = 9, y_4 = 16$$

$$\begin{aligned} \int_0^4 x^2 dx &= (1/3) [(0 + 16) + 4(1 + 9) + 2(4)] \\ &= (1/3) [16 + 4(10) + 8] \\ &= (1/3) [16 + 40 + 8] \\ &= (1/3) \times 64 \\ &= 21.333 \end{aligned}$$

Actual: 21.333 (Exact match for quadratic!)

4.7 Simpson's 3/8 Rule

Uses **cubic approximation**. Condition: **n must be MULTIPLE OF 3**.

Formula

$$\int_a^b f(x) dx = (3h/8) [y_0 + y_n + 3(y_1 + y_2 + y_4 + y_5 + \dots) + 2(y_3 + y_6 + \dots)]$$

Example: Find $\int_0^6 x^2 dx$

x	0	1	2	3	4	5	6
$y = x^2$	0	1	4	9	16	25	36

$a = 0, b = 6, n = 6$ (multiple of 3)

$h = (6 - 0) / 6 = 1$

$y_0 = 0, y_1 = 1, y_2 = 4, y_3 = 9$

$y_4 = 16, y_5 = 25, y_6 = 36$

$$\begin{aligned} \int_0^6 x^2 dx &= (3/8) [(0 + 36) + 3(1 + 4 + 16 + 25) + 2(9)] \\ &= (3/8) [36 + 3(46) + 18] \\ &= (3/8) [36 + 138 + 18] \\ &= (3/8) \times 192 \\ &= 72 \end{aligned}$$

Actual: $\int_0^6 x^2 dx = [x^3/3]_0^6 = 216/3 = 72$ (Exact!)

4.8 Weddle's Rule

Condition: n must be MULTIPLE OF 6.

Formula (for n = 6)

$$\int_a^b f(x) dx = (3h/10) [y_0 + y_6 + 5(y_1 + y_5) + 6(y_2 + y_4) + 2y_3]$$

Example: Find $\int_0^6 x^2 dx$

$$a = 0, b = 6, n = 6$$

$$h = (6 - 0) / 6 = 1$$

$$y_0 = 0, y_1 = 1, y_2 = 4, y_3 = 9$$

$$y_4 = 16, y_5 = 25, y_6 = 36$$

$$\begin{aligned} \int_0^6 x^2 dx &= (3/10)[0 + 36 + 5(1 + 25) + 6(4 + 16) + 2(9)] \\ &= (3/10)[36 + 5(26) + 6(20) + 18] \\ &= (3/10)[36 + 130 + 120 + 18] \\ &= (3/10) \times 304 \\ &= 91.2 \end{aligned}$$

Actual: 72

Error = 19.2 (Weddle's rule works best for higher degree polynomials)

4.9 Comparison of Integration Methods

For $\int_0^4 x^2 dx$ (Actual = 21.333)

Method	n	Result	Error	Accuracy
Trapezoidal	4	22.000	0.667	Low
Simpson's 1/3	4	21.333	0.000	High
Simpson's 3/8	3	21.333	0.000	High

Observations:

- Simpson's rules are more accurate for polynomials
- Trapezoidal rule gives overestimation
- More intervals = Better accuracy
- Weddle's rule best for higher degree polynomials

Formula Summary

Method	Condition	Formula
Trapezoidal	Any n	$(h/2)[y_0 + y_n + 2\sum y_i]$
Simpson's 1/3	n even	$(h/3)[y_0 + y_n + 4\sum y_{\text{odd}} + 2\sum y_{\text{even}}]$
Simpson's 3/8	n multiple of 3	$(3h/8)[y_0 + y_n + 3\sum y_{1,2,4,5\dots} + 2\sum y_{3,6\dots}]$
Weddle's	n multiple of 6	$(3h/10)[y_0 + y_6 + 5(y_1+y_5) + 6(y_2+y_4) + 2y_3]$

Accuracy Order



Trapezoidal < Simpson's 3/8 < Simpson's 1/3 < Weddle's

4.10 Previous Year Questions

Short Questions (2 marks)

1. What is numerical differentiation?
2. Write the trapezoidal rule formula.
3. What is the condition for Simpson's 1/3 rule?
4. What is the condition for Simpson's 3/8 rule?
5. What is Weddle's rule?
6. When is graphical differentiation used?
7. Write the central difference formula for derivative.
8. What is the difference between trapezoidal and Simpson's rule?

Medium Questions (5 marks)

1. Find $f'(2)$ from the data:
x: 0, 1, 2, 3, 4 y: 1, 2, 5, 10, 17
2. Evaluate $\int_0^4 (x^2 + 1) dx$ using trapezoidal rule with $h = 1$.
3. Evaluate $\int_0^6 x^2 dx$ using Simpson's 1/3 rule ($n = 6$).
4. Explain the forward difference formula for numerical differentiation.
5. Compare Simpson's 1/3 and Simpson's 3/8 rules.

Long Questions (12 marks)

1. (a) Explain numerical differentiation for equally spaced data with formulas and example.
(b) Find $f'(2)$ and $f''(2)$ from the data:
x: 0, 1, 2, 3, 4 y: 0, 1, 8, 27, 64

2. (a) Explain Newton-Cotes formulas for numerical integration.
 (b) Evaluate $\int_0^1 e^x dx$ using trapezoidal, Simpson's 1/3, and Simpson's 3/8 rules with $h = 0.25$.
3. (a) Compare trapezoidal, Simpson's 1/3, and Simpson's 3/8 rules with examples.
 (b) Evaluate $\int_0^6 \sqrt{x} dx$ using Weddle's rule with $h = 1$.

Quick Reference Card

Differentiation Formulas

Forward (at start)	$f'(x_0) = (1/h)[\Delta y_0 - \frac{1}{2}\Delta^2 y_0 + \frac{1}{3}\Delta^3 y_0 - \dots]$
Backward (at end)	$f'(x_n) = (1/h)[\nabla y_n + \frac{1}{2}\nabla^2 y_n + \frac{1}{3}\nabla^3 y_n + \dots]$
Central (interior)	$f'(x_i) = (y_{i+1} - y_{i-1}) / (2h)$
Second Derivative	$f''(x_i) = (y_{i-1} - 2y_i + y_{i+1}) / h^2$

Integration Formulas

Trapezoidal	$(h/2)[y_0 + y_n + 2\sum y_i]$
Simpson's 1/3	$(h/3)[y_0 + y_n + 4\sum y_{\text{odd}} + 2\sum y_{\text{even}}]$
Simpson's 3/8	$(3h/8)[y_0 + y_n + 3\sum y_{\text{group1}} + 2\sum y_{\text{group2}}]$
Weddle's	$(3h/10)[y_0 + y_6 + 5(y_1+y_5) + 6(y_2+y_4) + 2y_3]$